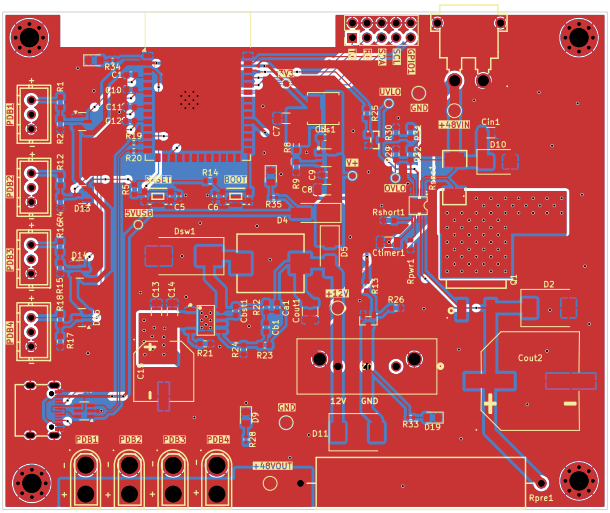


# Architecture Design, Sizing, and Vulnerability of the “Good Boy Limiter”

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GBL version 1

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A practical documentation for understanding the GBL architecture and sizing of elements.

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# 1 Abstract and System Overview

The design of high-power DC distribution networks frequently encounters the engineering challenge of managing massive capacitive loads during the initial charging phase. The "Good Boy Limiter" (GBL) represents an hybrid power management and in-rush current control architecture specifically designed to address these challenges within a 48 V DC ecosystem.

The main problem addressed by the GBL architecture is the safe charge and continuous operation of a downstream load characterized by a really large input capacitance. Specifically, the system must interface with a bulk bus capacitance ($C_{tot}$) of 7920 $\mu$F, which is physically realized through a distributed $4 \times 6$ matrix of 330 $\mu$F electrolytic capacitors. Furthermore, the system is designed to support a substantial nominal operating current of 260 A, although it is currently constrained to a 120 A limit for safety reasons.

Attempting to directly connect a 48 V stiff voltage source to an uncharged 7920 $\mu$F capacitor bank results in an instantaneous "in-rush" current theoretically limited only by the Equivalent Series Resistance (ESR) of the capacitors and the parasitic inductance of the interconnecting cables. In practical terms, this transient current can easily exceed several thousands of amperes. Such catastrophic current spikes induce severe electromagnetic interference (EMI), trigger upstream power supply overcurrent protections, and instantly weld the contacts of mechanical relays due to localized plasma arcing.

To resolve these conflicting requirements, the GBL employs a mixed relay/solid-state architecture. A heavy-duty mechanical external relay (actuated via a 12 V coil) serves as the primary conduction path during nominal steady-state operation. In parallel, a highly controlled solid-state pre-charge circuit, orchestrated by the Texas Instruments LM5069-2 hot-swap controller and an STMicroelectronics STB15810 MOSFET, governs the initial energization phase.

This report provides an exhaustive analysis of the pre-charge dynamics, focusing on the mathematical sizing of the LM5069-2 control parameters. It critically examines the thermal paradigm shift achieved by the inclusion of a series limiting resistor, assesses the Safe Operating Area (SOA) of the selected MOSFET, and documents the role of the ESP32-S3 microcontroller in mitigating topological safety vulnerabilities.

# 2 Power Architecture: Relay Versus Solid-State

High-current DC power distribution systems inherently suffer from the physical limitations of their switching elements. The architectural decision to utilize a hybrid system in the Good Boy Limiter is grounded in fundamental power dissipation physics.

## 2.1 The Limitations of Pure Solid-State Switching

In a hypothetical pure solid-state architecture managing the full 260 A nominal current, the conduction losses would be to high. Even utilizing state-of-the-art power MOSFETs exhibiting an ultra-low drain-to-source on-resistance ($R_{DS(on)}$) of 1 m$\Omega$, the continuous power dissipation ($P_{loss}$) would be substantial. According to Joule's law:

$$P_{loss} = I^2 \times R_{DS(on)} = (260 \text{ A})^2 \times 0.001 \ \Omega = 67.6 \text{ W} \tag{1}$$

Dissipating 67.6 W of pure thermal energy on a printed circuit board (PCB) requires massive copper planes, extensive thermal via arrays, large extruded aluminum heatsinks, and mandatory active forced-air cooling. This increases volumetric footprint, acoustic noise, and overall system failure rates.

### 2.2 The Limitations of Pure Mechanical Switching

Vice-versa, a purely mechanical relay architecture offers micro-ohm contact resistance, virtually eliminating steady-state thermal dissipation. However, mechanical relays are physically incapable of closing into an uncharged 7920 $\mu$F capacitive bus at 48 V. As the relay contacts physically approach one another during closure, the dielectric breakdown voltage of the microscopic air gap is exceeded. This initiates a high-energy plasma arc. The thousands of amperes of in-rush current flowing through this plasma instantly melt the contact plating, leading to contact welding and permanent failure of the relay.

### 2.3 The Hybrid Resolution

The GBL architecture resolves this dichotomy via a parallel power path topology:

- **The Solid-State Pre-Charge Path:** Engaged first, this path utilizes a series-pass N-channel MOSFET (STMicroelectronics STB15810) and a 50 Ω wirewound power resistor (EWWR0010J50R0T9) to strictly limit the transient charging current. The capacitors are charged safely over a calculated time domain.
- **The Main Relay Path:** Engaged only when the ESP32-S3 microcontroller verifies that the bus voltage has reached the nominal source voltage. Because the voltage differential ($\Delta V$) across the relay contacts is near zero, the corresponding current during closure is zero. Closing the relay under this Zero-Voltage Switching (ZVS) entirely eliminates arcing, preserving contact longevity. Once the relay is closed, the solid-state path is disabled.

## 3 Analysis and Sizing of the Pre-Charge Dynamics

The pre-charge network is mathematically modeled as a first-order ideal RC circuit. The primary system parameters dictate a DC source voltage ($V_{IN}$) of 48 V, a total bus capacitance ($C_{tot}$) of 7920 $\mu$F, and a physical current-limiting series resistance ($R_{pre}$) of 50 Ω.

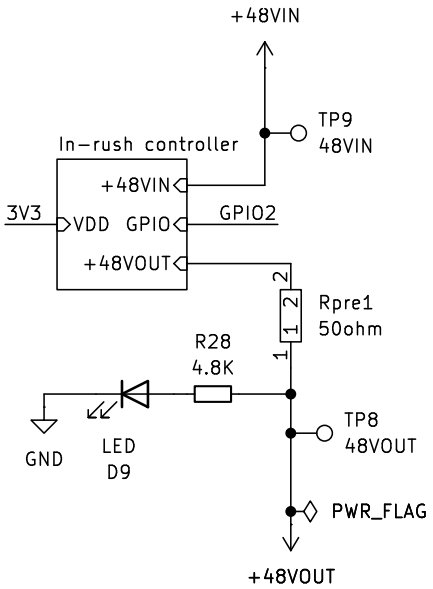



Figure 1: Detail of the solid-state pre-charge topology. The 50 Ω resistor $R_{PRE1}$ dominates the branch impedance, physically limiting the maximum current to 0.96 A regardless of the state of the LM5069 controller.

### 3.1 Peak In-Rush Current Constraint

The theoretical peak current ($I_{peak}$) occurs instantaneously at $t = 0$, assuming the capacitor bank is fully discharged ($V_C = 0$ V). The peak current is defined by Ohm's Law across the series resistor:

$$I_{peak} = \frac{V_{IN}}{R_{pre}} = \frac{48 \text{ V}}{50 \ \Omega} = 0.96 \text{ A} \tag{2}$$

By actively restricting the peak current to a mere 0.96 A, the GBL architecture completely eliminates the stress on the upstream 48 V power supply. The upstream source perceives the startup event as a negligible, highly controlled 46 W load rather than a multi-kilowatt short circuit.

### 3.2 Time Constant and RC Charging Profile

The exponential voltage charging profile of the capacitor bank is governed by the RC time constant ($\tau$), which dictates the temporal response of the circuit:

$$\tau = R_{pre1} \times C_{tot} = 50 \ \Omega \times 0.00792 \text{ F} = 0.396 \text{ s} \tag{3}$$

The bus voltage as a function of time, $V_C(t)$, is defined by the standard step-response differential equation solution:

$$V_C(t) = V_{IN} \left(1 - e^{\frac{-t}{\tau}}\right) = 48 \left(1 - e^{\frac{-t}{0.396}}\right) \tag{4}$$

In standard engineering practice, a capacitor is considered functionally charged at $3\tau$ (95.0% of $V_{IN}$) and fully charged at $5\tau$ (99.3% of $V_{IN}$).

We can calculate the exact time required to reach the 95% threshold (45.6 V):

$$t_{95\%} = 3 \times \tau = 3 \times 0.396 \text{ s} = 1.188 \text{ s} \tag{5}$$

Furthermore, the time required to reach the 99% threshold (47.5 V):

$$t_{99\%} = 5 \times \tau = 5 \times 0.396 \text{ s} = 1.980 \text{ s} \tag{6}$$

These timing calculations define the minimum operational window required before the ESP32-S3 firmware can safely command the main mechanical relay to close. Actuating the relay before 1.188 s would risk residual arcing due to a significant remaining voltage differential.

### 3.3 Energy Transfer Thermodynamics

The total energy transferred from the 48 V source to fully charge the capacitor bank is derived from the energy stored in the electric field of the capacitors:

$$E_{cap} = \frac{1}{2} C_{tot} V_{IN}^2 = \frac{1}{2} \times 0.00792 \text{ F} \times (48 \text{ V})^2 = 9.1238 \text{ J} \tag{7}$$

A fundamental principle of thermodynamics in RC charging circuits dictates that an exactly equal amount of energy is dissipated as heat within the resistive elements of the charging path. Therefore, regardless of the charging speed, exactly 9.12 J of thermal energy is deposited into the 50 Ω wire-wound resistor during every startup sequence. This will be critically analyzed in Section 6.

## 4 In-Rush Controller (LM5069) Analysis and Sizing

The Texas Instruments LM5069-2 operates as the gate driver and primary hardware safety controller for the solid-state pre-charge branch.

The following subsections detail the mathematical sizing of its internal thresholds.

### 4.1 Undervoltage Lockout (UVLO) Sizing

The UVLO pin ensures the controller only activates when the input bus is stable and possesses sufficient voltage to properly drive the downstream loads. The LM5069 possesses a precise internal comparator with an enable threshold of 2.5 V. To prevent rapid ON/OFF oscillation (chattering) when the input voltage fluctuates near the threshold, the IC utilizes an internal 21 $\mu$A current source to provide a programmable hysteresis.

The GBL schematic defines a dedicated UVLO resistive divider utilizing $R_{25} = 143$ k$\Omega$ as the top resistor and $R_{30} = 10$ k$\Omega$ as the bottom resistor connected to ground.

The rising input threshold voltage ($V_{\text{UVLO_RISE}}$) at which the controller enables the pre-charge sequence is calculated as:

$$V_{\text{UVLO_RISE}} = V_{REF} \times \frac{R_{25} + R_{30}}{R_{30}} = 2.5 \text{ V} \times \frac{143 \text{ k}\Omega + 10 \text{ k}\Omega}{10 \text{ k}\Omega} = 38.25 \text{ V} \tag{8}$$

Once the input voltage exceeds 38.25 V, the 21 $\mu$A internal current source activates and flows out of the UVLO pin, through the bottom resistor $R_{30}$, effectively raising the pin voltage. This creates the hysteresis voltage ($V_{\text{UVLO_HYS}}$), which is dictated by the current source acting upon the top resistor $R_{25}$:

$$V_{\text{UVLO_HYS}} = I_{HYS} \times R_{25} = 21 \ \mu\text{A} \times 143 \text{ k}\Omega = 3.003 \text{ V} \tag{9}$$

The falling threshold voltage ($V_{\text{UVLO_FALL}}$), where the controller will abort the sequence due to a sagging input bus, is therefore:

$$V_{\text{UVLO_FALL}} = V_{\text{UVLO_RISE}} - V_{\text{UVLO_HYS}} = 38.25 \text{ V} - 3.003 \text{ V} = 35.247 \text{ V} \tag{10}$$

Therefore, the pre-charge sequence is hardware-locked until the input voltage is a steady 38.25 V, and it possesses a robust 3 V noise margin before shutting down at 35.24 V.

### 4.2 Overvoltage Lockout (OVLO) Sizing

Similarly, the OVLO pin protects the downstream systems from catastrophic input voltage surges, such as load-dump events or incorrect power supply connections. To maintain precise and independent control over the overvoltage threshold, the GBL schematic utilizes a separate, dedicated divider string: $R_{32} = 210$ k$\Omega$ as the top resistor and $R_{29} = 10$ k$\Omega$ as the bottom resistor.

The rising overvoltage cutoff threshold ($V_{\text{OVLO_RISE}}$) is calculated using the same 2.5 V internal reference:

$$V_{\text{OVLO_RISE}} = V_{REF} \times \frac{R_{32} + R_{29}}{R_{29}} = 2.5 \text{ V} \times \frac{210 \text{ k}\Omega + 10 \text{ k}\Omega}{10 \text{ k}\Omega} = 55.00 \text{ V} \tag{11}$$

The OVLO pin also features a 21 $\mu$A hysteresis current source. The hysteresis voltage is:

$$V_{\text{OVLO_HYS}} = I_{HYS} \times R_{32} = 21 \ \mu\text{A} \times 210 \text{ k}\Omega = 4.410 \text{ V} \tag{12}$$

The falling recovery threshold ($V_{\text{OVLO_FALL}}$), where the system permits operation to resume after an overvoltage event clears, is:

$$V_{\text{OVLO_FALL}} = V_{\text{OVLO_RISE}} - V_{\text{OVLO_HYS}} = 55.00 \text{ V} - 4.41 \text{ V} = 50.59 \text{ V} \tag{13}$$

The hardware overvoltage cutoff is situated at 55.00 V. This provides adequate operational margin above the nominal 48 V state, while strictly protecting downstream components from transient degradation.

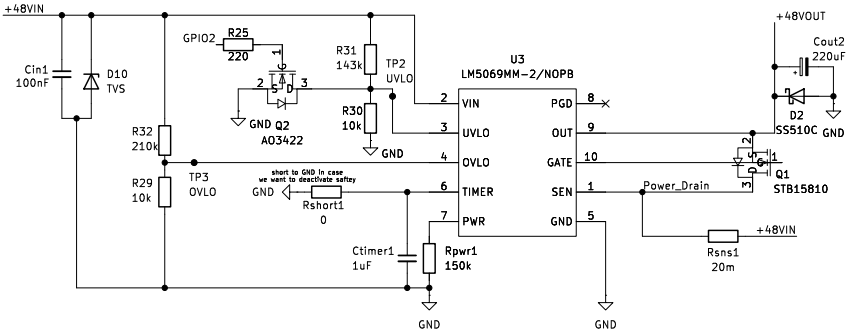



Figure 2: Schematic of the independent resistive dividers for the undervoltage ($R_{25}$, $R_{30}$) and overvoltage ($R_{32}$, $R_{29}$) thresholds. The use of two separate networks ensures clear and independent hysteresis for the operating windows at 38.25 V and 55.00 V.

### 4.3 Hardware Current Limit ($I_{LIM}$) and $R_{sns1}$ Sizing

The standard operational mode of the LM5069 relies on sensing the voltage drop across a precision shunt resistor ($R_{SENS1}$) placed directly between the **VIN** and **SENSE** pins. If the load current increases such that the voltage across the sense resistor reaches 55 mV, the LM5069 limits the current by dynamically adjusting the MOSFET gate voltage.

In the GBL design, a value for $R_{SENS1} = 20$ m$\Omega$ is utilized. The absolute hardware current limit is established mathematically as:

$$I_{LIM} = \frac{V_{SNS(th)}}{R_{SENS1}} = \frac{55 \text{ mV}}{20 \text{ m}\Omega} = 2.75 \text{ A} \tag{14}$$

### 4.4 Power Limit Sizing ($P_{FET(LIM)}$)

To protect the external MOSFET from violating its Safe Operating Area (SOA) during linear mode regulation, the LM5069 allows the maximum allowable power dissipation of the FET to be programmed via the PWR pin. The LM5069 monitors the $V_{DS}$ of the MOSFET by comparing the SENSE pin and the OUT pin.

This limit is set using an external resistor ($R_{PWR1}$) connected to the PWR pin. The schematic indicates $R_{PWR1} = 150$ k$\Omega$.

The formula correlating the programmable power limit to the sense resistor and the PWR resistor is provided by Texas Instruments as:

$$P_{FET(LIM)} = \frac{R_{PWR1}}{1.25 \times 10^5 \times R_{SENS1}} = \frac{150,000 \ \Omega}{1.25 \times 10^5 \times 0.020 \ \Omega} = 60 \text{ W} \tag{15}$$

The hardware is theoretically configured to throttle the MOSFET gate if it detects the MOSFET is dissipating more than 60 W of heat. However, the presence of the 50 $\Omega$ resistor ensures the MOSFET never dissipates anywhere near 60 W. This will be proven in Section 5.

### 4.5 Fault Timer Duration ($t_{FAULT}$)

The fault timeout allows the system to ride through brief, acceptable transient over-currents without nuisance tripping, but ensures the MOSFET is definitively shut down if a fault persists long enough to threaten the system. The timing is dictated by an external capacitor $C_{TIMER1} = 1$ $\mu$F connected to the TIMER pin.

The typical duration before a fault is declared and the gate is pulled low is defined by the equation:

$$t_{FAULT} = C_{TIMER1} \times \frac{v_{timer,typ}}{i_{timer,typ}} = 1 \times 10^{-6} \text{ F} \times 4.7 \times 10^4 = 0.047 \text{ s} = 47 \text{ ms} \tag{16}$$

The specific LM5069-2 variant utilized in this design features an Auto-Retry mechanism. If the controller were to enter a current or power limit regulation state for longer than the 47 ms timer, the device would shut down the MOSFET, run through an extended internal cool-down cycle (dictated by the same capacitor), and then attempt to restart the process automatically. However, as established, the controller is physically prevented from ever entering the regulation state due to the 50 Ω resistor.

# 5 Thermal Management and MOSFET Safe Operating Area (SOA)

The defining architectural characteristic of the GBL pre-charge circuit is the intentional shifting of the massive thermal dissipation burden away from the semiconductor lattice of the STB15810 MOSFET and onto the robust, passive wirewound resistor.

## 5.1 The STB15810 MOSFET in Deep Saturation

In the Good Boy Limiter architecture, the presence of the 50 Ω resistor physically restricts the maximum current to 0.96 A. Because 0.96 A is vastly lower than the LM5069's 2.75 A current regulation threshold, the LM5069's internal high-side charge pump continues to pump unabated. It fully enhances the MOSFET gate, raising $V_{GS}$ to its maximum value (typically 10 V to 12 V above the source). This pushes the MOSFET completely out of the linear region and deep into the ohmic region (saturation).

The STB15810 is a heavily over-specified 100 V, 110 A device utilizing advanced STripFET™ F7 trench gate technology, boasting an ultra-low $R_{DS(on)}$ of 3.9 mΩ typical and 7 mΩ absolute maximum.

Assuming the worst-case scenario at maximum operating junction temperature where $R_{DS(on)} = 7$ mΩ, the peak power dissipated by the MOSFET during the absolute highest current flow of the pre-charge phase (0.96 A at $t = 0$) is simply calculated using Joule's law for a pure resistance:

$$P_{MOSFET(peak)} = I_{peak}^2 \times R_{DS(on)} = (0.96 \text{ A})^2 \times 0.007 \ \Omega = 0.0064 \text{ W} = 6.4 \text{ mW} \tag{17}$$

Dissipating a mere 6.4 mW means the STB15810 will experience absolutely zero measurable thermal rise above ambient temperature. The device remains entirely safe, utterly immune to the Spirito effect, and operates infinitely deep within the safest bounds of its DC SOA curve. From the perspective of thermal stress, the MOSFET acts as a perfect wire.

# 6 Thermal Limitations: Analyzing the 50 Ω Power Resistor

Because the MOSFET acts as a near-perfect conductor, the 50 Ω wirewound resistor assumes basically 100% of the thermal load during the pre-charge event.

During the exact moment of turn-on ($t = 0$), the capacitor bank acts as a short circuit, meaning the entire 48 V source is dropped across the resistor. The instantaneous peak power dissipation is:

$$P_{RESISTOR(peak)} = \frac{V_{IN}^2}{R_{pre}} = \frac{(48 \text{ V})^2}{50 \ \Omega} = 46.08 \text{ W} \tag{18}$$

The selected component, EWWR0010J50R0T9, is a wirewound power resistor rated for 10 W of continuous steady-state power. Operating a 10 W rated resistor at 46.08 W could appears as a design error; however, it is permissible because pre-charging is a highly transient pulse event, not a steady-state condition.

As calculated in Equation (7), the total energy deposited into the resistor over the entire 1.5 to 2.0 second charging curve is strictly bounded to 9.12 J. Wirewound resistors are specifically

constructed by winding a resistive alloy wire around a dense ceramic core. This ceramic core possesses a significant thermal mass. During a short 2-second pulse, the heating is largely adiabatic, the energy is absorbed directly into the thermal mass of the ceramic core, resulting in a moderate, entirely safe transient temperature rise.

The critical operational limitation here is duty cycle: the system must not be commanded to repeatedly toggle the pre-charge sequence in rapid succession. Sufficient cool-down time (several seconds) must be enforced by the ESP32 software to allow the ceramic core to shed this thermal energy to the ambient environment.

# 7 Control Logic, System Security, and the Software Timeout

While the delegation of the current-limiting role to a passive 50 Ω resistor safely protects the MOSFET, it simultaneously creates a critical vulnerability within the hardware protection layers of the LM5069. This vulnerability necessitates the implementation of an active software safety supervisory system via the ESP32-S3 microcontroller.

## 7.1 The "Hardware Blind Spot" during a Hard Short

Consider a catastrophic operational scenario: a dead short circuit directly to ground exists on the 48 V output bus (e.g., a crushed cable or a failed downstream component) before the system is turned on.

When the ESP32 initiates the pre-charge sequence, the 50 Ω resistor performs its physical duty and limits the fault current:

$$I_{fault} = \frac{48 \text{ V}}{50 \ \Omega + 0.020 \ \Omega + 0.007 \ \Omega} \approx 0.96 \text{ A} \tag{19}$$

Because the fault current (0.96 A) is significantly lower than the LM5069 hardware current limit threshold (2.75 A), the LM5069 internal current comparator is never triggered. Furthermore, because the 50 Ω resistor is dropping nearly the entire 48 V, the voltage at the SENSE pin is roughly 0 V, and the $V_{DS}$ across the MOSFET is negligible (a few millivolts). Consequently, the power limit comparator ($P_{LIM} = 60$ W) is also completely blind to the fault.

**The Catastrophic Result:** The LM5069 views the dead short circuit as a completely normal, healthy operating condition. The internal 47 ms fault timer ($C_{TIMER}$) never begins counting. The hardware safety system will remain engaged indefinitely, forcing the 10 W wire-wound resistor to continuously dissipate 46.08 W. Within a matter of seconds, the resistor will violently exceed its thermal degradation limits, resulting in extreme overheating, outgassing, potential ignition of surrounding components, and ultimately an open-circuit failure.

## 7.2 The ESP32-S3 Software Timeout Mitigation

To permanently resolve this dangerous topological vulnerability, the ESP32-S3 microcontroller is tasked with monitoring the dynamic health of the pre-charge process via the dedicated Voltage Sensing Modules.

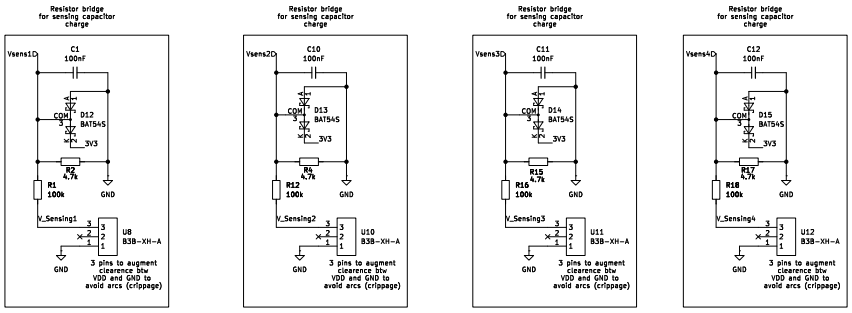



Figure 3: Schematic diagram of the V_Sensing modules used by the ESP32 to read the bus voltage. Note the resistor dividers connected to the BAT54S diodes, which are essential for protecting the microcontroller's ADC inputs from transient overvoltages.

The ESP32 reads the intermediate bus voltage via a resistor divider string ($R_{14} = 10$ k$\Omega$, etc.). These inputs are aggressively protected by dual BAT54S Schottky clamping diodes, which prevent the microcontroller's sensitive ADC pins from exceeding the 3.3 V logic supply limit even during severe transients.

The ESP32 firmware must be programmed to implement a strict, non-blocking state-machine timeout logic:

1. **Initiation:** The ESP32 asserts the LM5069 EN/UVLO pin HIGH via a GPIO pin to begin the pre-charge sequence. Simultaneously, it initializes an internal hardware software timer ($t = 0$).

2. **Continuous Monitoring:** The ADC continuously samples the bus voltage ($V_{bus}$) at a high frequency.

3. **Success Condition:** If $V_{bus}$ successfully exhibits the standard RC charging curve and exceeds 45 V (approximately $3\tau$) within a 1.5 to 2.0 second window, the pre-charge is deemed successful. The ESP32 then commands the main 12 V mechanical relay to close, routing the 260 A capability to the load. Once the relay confirms closure, the solid-state branch is deactivated.

4. **Timeout Failure Condition (The Critical Mitigation):** If $V_{bus}$ fails to rise above a minimum safety threshold within 3.0 seconds (a "Timeout of Security"), it is definitive proof of a downstream fault or short circuit absorbing the charge current. The ESP32 must immediately and unconditionally force the EN/UVLO pin LOW. This action forcefully shuts down the LM5069, halting the current flow and saving the 50 $\Omega$ resistor from thermal destruction.

This software-defined safety boundary bridges the gap left by the hardware topology, ensuring full system resilience against all modes of failure.

# 8 Auxiliary Power Supply: LM5013 DC-DC Converter

To drive the heavy inductive coil of the main mechanical relay and provide logic-level power to the digital supervision systems, the GBL integrates a high-voltage auxiliary step-down (buck) converter based on the Texas Instruments LM5013DDAR.

The LM5013 steps the primary 48 V distribution bus down to an intermediate 12 V rail. The converter is capable of handling inputs up to 100 V, providing massive immunity against 48 V bus ringing and transients. It operates utilizing a Constant On-Time (COT) architecture. COT control provides exceptional transient load response (essential when slamming a heavy

relay coil) and entirely eliminates the need for complex external loop compensation networks (type II or type III compensators), simplifying the PCB layout.

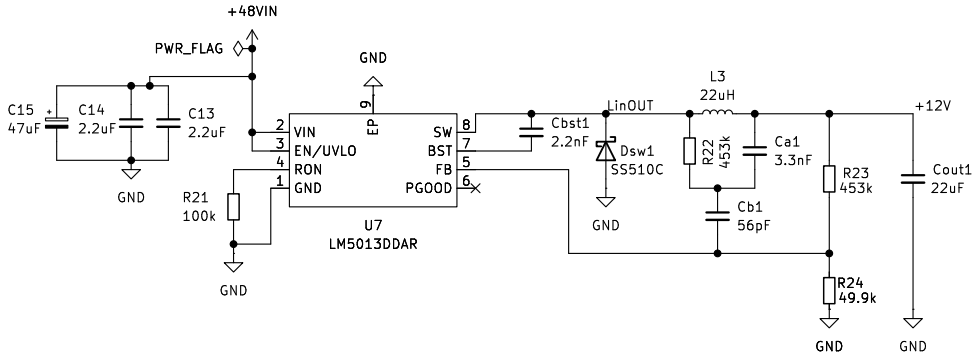



Figure 4: Topology of the LM5013 step-down converter. The main components are: the 22 $\mu$H inductor $L_3$, the bypass diode Dsw1 (SS510C), and the $R_{23}/R_{24}$ feedback network for precise regulation of the 12 V output.

### 8.1 Feedback Sizing for the 12 V Rail

The output voltage is programmed via a precision resistor divider feeding back into the FB pin. The highly precise internal bandgap reference voltage of the LM5013 is exactly $V_{REF} = 1.2$ V. The schematic defines the top feedback resistor as $R_{FB1}(R_{23}) = 453$ k$\Omega$ and the bottom resistor connected to ground as $R_{FB2}(R_{24}) = 49.9$ k$\Omega$.

The nominal output voltage is mathematically derived as:

$$V_{OUT} = V_{REF} \times \left(1 + \frac{R_{FB1}}{R_{FB2}}\right) = 1.2 \text{ V} \times \left(1 + \frac{453 \text{ k}\Omega}{49.9 \text{ k}\Omega}\right) \tag{20}$$

$$V_{OUT} = 1.2 \text{ V} \times (1 + 9.078) = 1.2 \text{ V} \times 10.078 = 12.09 \text{ V} \tag{21}$$

### 8.2 Core Buck Components and Filtering

The power stage utilizes a low-DCR inductor $L_3 = 22$ $\mu$H and an SS510C Schottky freewheeling diode ($D_{SW1}$). The presence of the external Schottky diode is mandatory for this non-synchronous topology to provide a path for the inductor current during the MOSFET off-time. The Constant On-Time programming resistor is $R_{ON}(R_{21}) = 100$ k$\Omega$, establishing the nominal switching frequency.

Input voltage ripple and filtering are aggressively handled by dual low-ESR 2.2 $\mu$F ceramic capacitors ($C_{13}, C_{14}$) plus a larger 47 $\mu$F bulk capacitor ($C_{15}$). This network attenuates high-frequency switching noise, preventing the buck converter from injecting harmonic distortions back into the main 48 V distribution bus. A secondary step-down stage (utilizing the SY8120B buck converter) subsequently drops this intermediate 12 V to a highly regulated 3.3 V rail to power the ESP32-S3 microcontroller and its associated sensitive analog peripherals.

## 9 Power Supply Logic and 3V3 Selector (SY8120B and Case Control)

The ESP32-S3 and the analog logic arrays require a stable, low-noise 3.3V power rail. To ensure the microcontroller remains active and functional under multiple operating conditions, such as during initial programming via a PC (USB power), standalone operation from external logic

batteries, or standard nominal operation connected to the main 48V bus, a specialized "3V3 Selector" architecture is implemented.

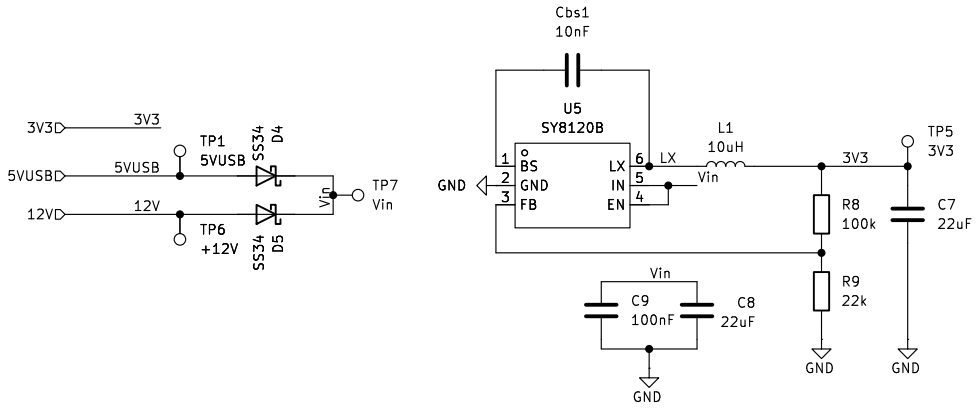



Figure 5: Schematic diagram of the "3V3 Selector" module based on the SY8120B synchronous buck converter. Note the OR-gate diode configurations (D4 and D5) for transparent switching between the USB power supply and the 12V bus, and the R10/R11 feedback divider for precise regulation to 3.3V.

## 9.1 Power Source OR-ing Network and Operational Scenarios

The input to the 3.3V regulation stage ($V_{in}$) is sourced dynamically via a passive diode OR-ing network. This network prevents reverse-current back-feeding and allows the system to autonomously draw power from the most viable source without logic interruption. The system relies on two SS34 Schottky diodes (D4 and D5) which feature a low forward voltage drop ($V_f \approx 0.4$V at nominal logic currents).The architecture supports three distinct operational scenarios:

- **Case 1: PC Configuration Only (No Main Battery Connected).**
  When the GBL is completely disconnected from the 48V high-voltage bus (no XT30 battery connected) and plugged into a host PC via the USB-C connector, the cable supplies $+5\text{V}_{USB}$. Diode D4 becomes forward-biased. The internal node $V_{in}$ reaches approximately 4.6V ($5\text{V} - V_{f(D4)}$). Because the SY8120B regulator has a minimum operational input threshold of 4.0V, it successfully activates, generating the 3.3V rail. This allows the ESP32 to be programmed, flashed, or debugged safely on a workbench without exposing the user to high-voltage hazards.

- **Case 2: Main Battery / 48V Bus Nominally Connected.**
  During standard operation, the main power source (e.g., a 48V battery pack) is connected to the XT30 connectors (CN1). The upstream LM5013 buck converter actively steps the 48V down to the intermediate +12V rail. This 12V potential reaches the anode of diode D5. D5 becomes forward-biased, pulling the $V_{in}$ node to approximately 11.6V ($12\text{V} - V_{f(D5)}$). The SY8120B easily accepts this input (rated up to 18V maximum) and drops it to 3.3V.

- **Case 3: Both Main Battery and PC USB Connected.**
  This is the most critical cross-power scenario. If a user connects the USB-C cable for telemetry while the 48V bus is fully live, the OR-ing network automatically resolves the power conflict. The anode of D5 sits at +12V, while the anode of D4 sits at $+5\text{V}_{USB}$. Because the cathodes of both diodes are tied together at the $V_{in}$ node, D5 will strongly conduct and pull the $V_{in}$ node up to 11.6V. Consequently, the +11.6V at the cathode of

D4 is significantly higher than the +5V at its anode. Diode D4 becomes heavily reverse-biased and stops conducting completely.

**Result:** The logic array draws zero power from the host PC's USB port, protecting the laptop's motherboard and ensuring the ESP32 is powered exclusively by the main battery line, while keeping the communication lines active for interfacing the MCU.

### 9.2 SY8120B Synchronous Buck Converter Sizing

Unlike the LM5013, the SY8120B is a *synchronous* buck converter. It integrates both the high-side and low-side switching MOSFETs internally. These internal switches boast very low $R_{DS(on)}$ characteristics (145 mΩ top / 135 mΩ bottom), minimizing conduction losses and **entirely eliminating the need for an external Schottky freewheeling diode**.

The output voltage is programmed by the external resistor divider connected to the feedback (FB) pin. The internal bandgap reference voltage of the SY8120B is $V_{REF} = 0.6$V.Utilizing the designated top resistor $R_{10} = 100$ kΩ and the bottom resistor $R_{11} = 22$ kΩ, the nominal output voltage is derived using the standard feedback formula:

$$V_{OUT} = V_{REF} \times \left(1 + \frac{R_{10}}{R_{11}}\right) = 0.6 \text{ V} \times \left(1 + \frac{100 \text{ k}\Omega}{22 \text{ k}\Omega}\right) = 3.327 \text{ V} \tag{22}$$

The calculated 3.327V output provides a regulated logic-level rail for the ESP32-S3. An external pull-up resistor $R_8 = 100$ kΩ connects the EN (Enable) pin directly to the input voltage node ($V_{in}$), ensuring the regulator automatically initiates switching the millisecond a valid power source (USB or Battery) is applied via the OR-ing networkr.

## 10 Main Relay Actuation and Control Circuit

The primary high-current conduction path of the GBL architecture relies on an external heavy-duty mechanical relay connected via U2 (TE Connectivity 1-178136-3). To interface the 3.3 V logic domain of the ESP32-S3 with the 12 V DC requirement of the relay coil, a dedicated low-side switching circuit is implemented on the PCB.

### 10.1 Low-Side Switching Topology

The relay coil is permanently tied to the +12 V auxiliary power rail at pin 3 of connector U2. The ground return path (pin 2) is dynamically controlled by an N-channel logic-level MOSFET, Q4 (AO3400A).

When the microcontroller asserts the `GPIO38` signal HIGH (3.3 V), Q4 enters conduction, pulling pin 2 to ground and energizing the relay coil. A 220 Ω series gate resistor (R13) limits the instantaneous transient current required to charge the MOSFET gate capacitance, protecting the ESP32 GPIO pin from overcurrent spikes. Furthermore, a 10 kΩ pull-down resistor (R26) ensures the gate is firmly held at 0 V during system boot or whenever the GPIO pin is in a high-impedance (floating) state, preventing unintended and dangerous relay activation.

### 10.2 Inductive Transient Protection

Due to the highly inductive nature of the relay coil, abruptly interrupting the coil current generates a massive reverse-voltage spike (back-EMF) as the internal magnetic field collapses. To protect the AO3400A MOSFET from exceeding its drain-to-source breakdown voltage, a Schottky flyback diode, D11 (SS510C), is placed in anti-parallel with the coil terminals (cathode connected to +12 V, anode to the MOSFET drain). This topology provides a safe, low-impedance freewheeling path to dissipate the inductive energy safely upon shutdown.

### 10.3 Design Anomaly: LED Indicator Inversion and Parasitic Draw

**Hardware Warning:** An architectural anomaly exists within the visual status signaling subcircuit associated with the relay. The LED D19 and its current-limiting resistor R33 (1 kΩ) are connected directly between the MOSFET drain (relay pin 2) and ground.

This topology results in an inverted logic state and an unintended continuous parasitic current draw:

- **Relay OFF State (Q4 Open):** The +12 V potential traverses the relay coil, passing out of pin 2, through R33 and D19, to ground. This actively illuminates the LED when the relay is *inactive*. Additionally, this establishes a continuous parasitic current of approximately 10 to 12 mA through the relay coil. While this current is insufficient to mechanically actuate the heavy relay contacts, it results in unnecessary thermal dissipation within the coil and constant power consumption from the 12 V rail.
- **Relay ON State (Q4 Closed):** The MOSFET pulls the drain to 0 V, eliminating the potential difference across the LED branch. As a result, the LED turns *off* when the relay is successfully engaged, acting inversely to standard visual indication practices.

**Mitigation Strategy:** For future PCB revisions, the LED branch should be relocated to the logic side (driven directly by `GPIO38` to ground) or placed strictly in parallel with the coil. For the current hardware iteration, if the 12 mA parasitic draw is deemed unacceptable for battery operation, it is recommended to physically desolder and remove R33 from the PCB. This will open the parasitic circuit loop, accepting the loss of visual status indication in favor of strict electrical efficiency.

## 11 LEDs

### 11.1 LED's Resistors Sizing

To light an LED from a power rail, you simply place the LED in series with a current-limiting resistor across the rail and ground. You must calculate the correct resistance ($R$) using Ohm's Law for preventing the LED to burn out:

$$R = \frac{V_{rail} - V_f}{I_f} \tag{23}$$

- $V_{rail}$: The voltage of your power rail (e.g., 3.3V, 5V, 12V).
- $V_f$: The Forward Voltage of the LED (typically 1.8V for Red, 3.2V for Blue/White).
- $I_f$: The Forward Current you want (typically 2mA to 20mA).

### 11.2 GBL's LEDs Color Code

- **RED**: +48V Pre-charge power rail is ON
- **YELLOW**: +3V3 power rail is ON
- **GREEN**: Relay is closed
- **BLUE**: +5USB power rail is ON
- **ORANGE**: +12V rail is ON

# 12 Conclusion and Recommendations

The "Good Boy Limiter" architecture represents a possible solution to the challenge of managing massive capacitive loads in high-current DC environments. By utilizing a mixed-topology parallel design, it combines the zero-loss continuous conduction capabilities of a mechanical relay with the non-destructive startup characteristics of a solid-state pre-charge circuit.

The sizing of the Texas Instruments LM5069-2 components is mathematically made. The selected resistor dividers establish 38.25 V and 55.00 V operational voltage lockouts, ensuring the system only operates within a safe 48 V nominal window. The absolute hardware current limit is determinate by $R_{SENS1}$ at 2.75 A.

Furthermore, the decision to utilize a 50 Ω series power resistor shields the STB15810 MOSFET from the destructive Safe Operating Area (SOA) stresses typically associated with linear-mode hot-swap events. By pushing the MOSFET deep into saturation, its thermal load is reduced to a negligible 6.4 mW, transferring the energy entirely to the adiabatic mass of the wirewound resistor.

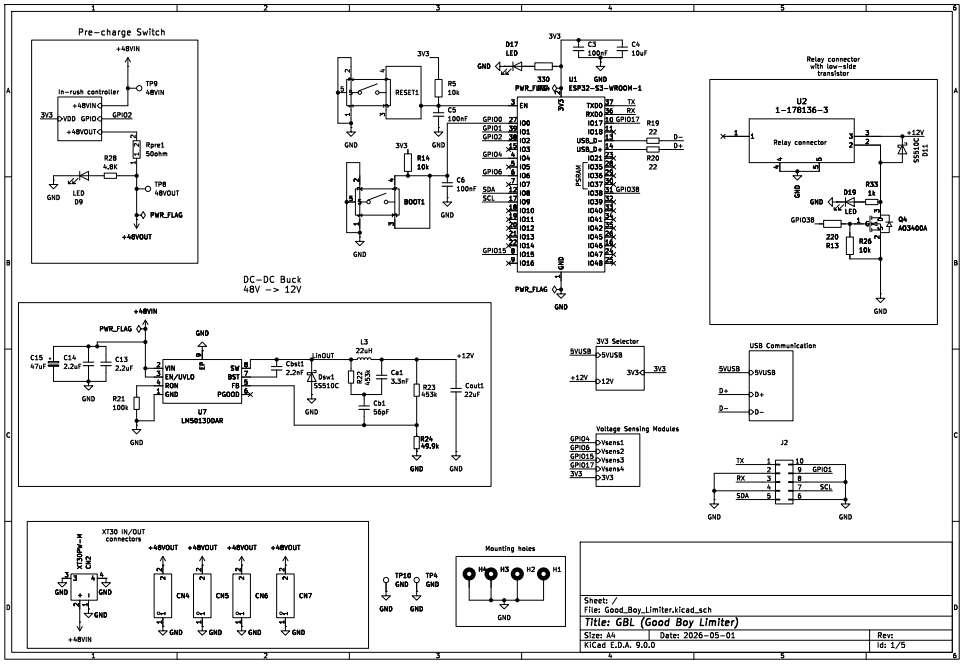
**Therefore, it is an absolute, critical project requirement to implement the 3.0 second Software Safety Timeout logic on the ESP32-S3**. Without this digital supervisor actively pulling the LM5069 EN pin low upon a failure to charge, the 10 W pre-charge resistor will face inevitable and thermal failure under a continuous 46 W fault load.

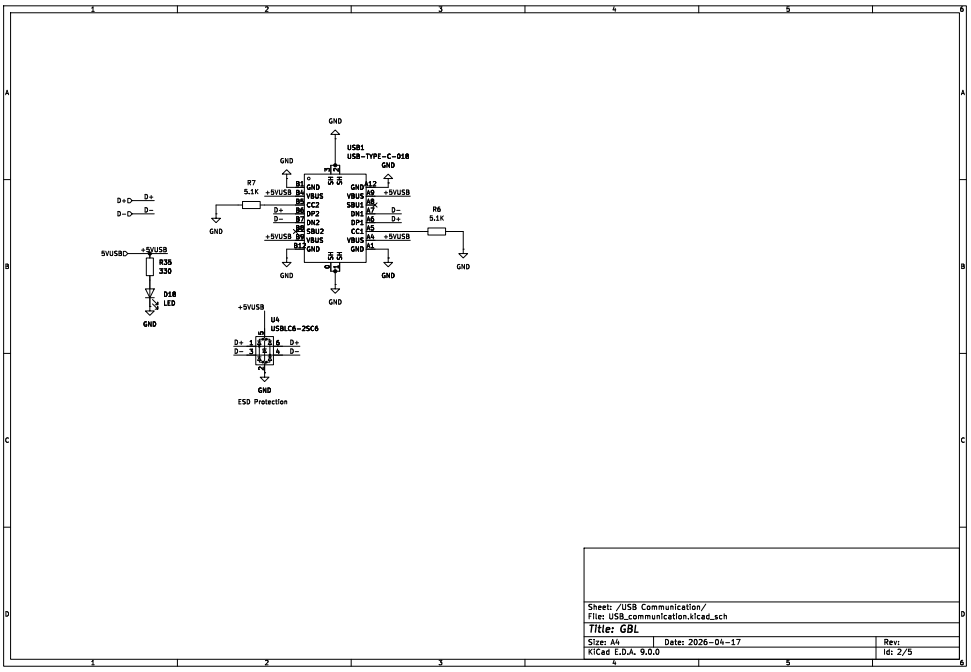
# 13 Next Steps for future iterations

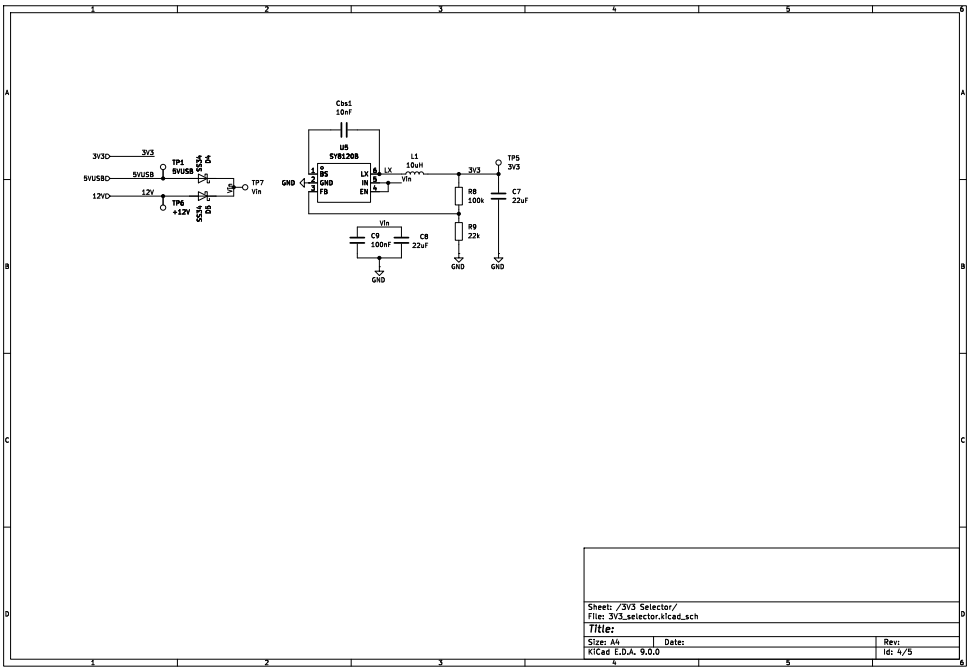
- Sizing optimization
- Design a better Solid-State branch
  - Better LM5069 safety features
  - Charge pump (at least to 48V + $V_{GS(th)}$) with $V_{GS} = 48V$ SS
  - Gate driver with 48V + $V_{GS(th)}$ and boost converter
- Routing optimization

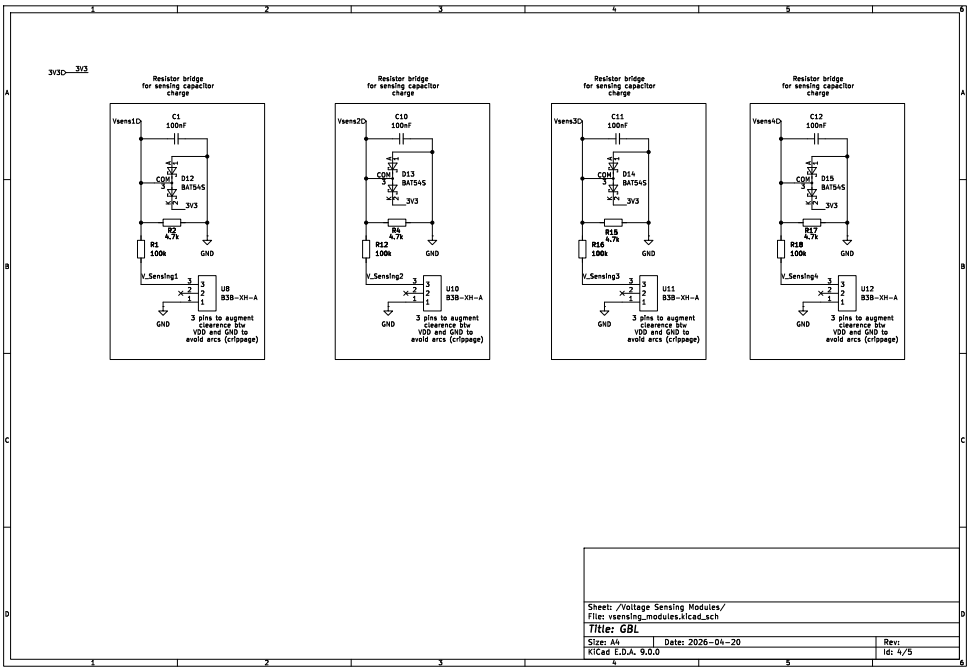
# 14 Annexes

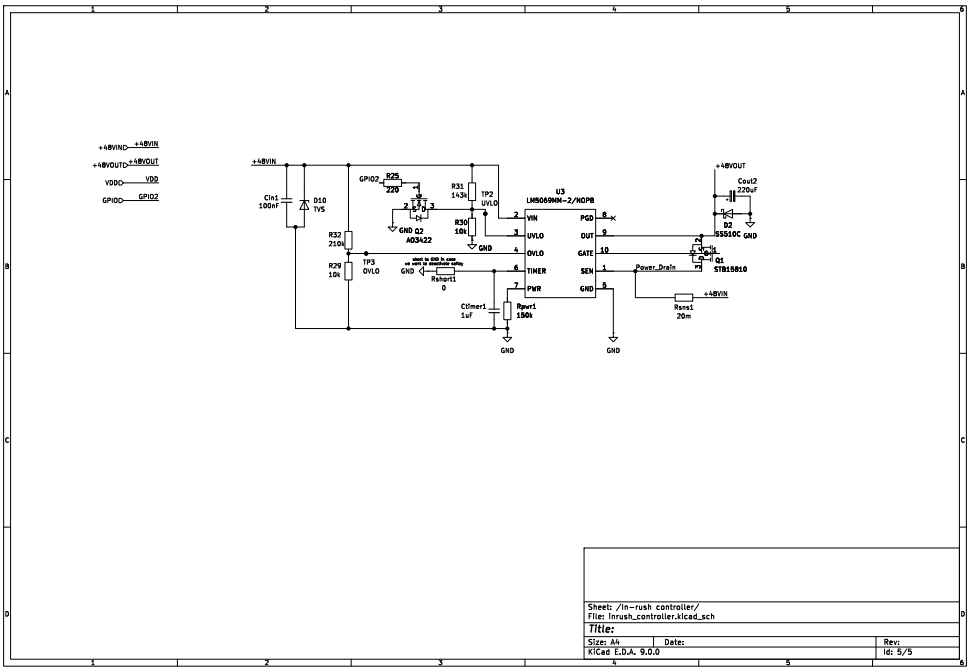
Complete GBL e-scheme:


Pre-charge Switch
+48VIN
TP9
48VIN
In-rush controller
+48VIN
3V3
VDD
GPIO
GPIO2
+48VOUT
Rpre1
50ohm
R28
4.8K
LED
D9
TP8
48VOUT
PWR_FLAG
+48VOUT
DC-DC Buck
48V -> 12V
PWR_FLAG
+48VIN
GND
C15
47uF
C14
2.2uF
C13
2.2uF
VIN
EN/UVLO
RON
GND
SW
BST
FB
PGOOD
EP
U7
LM5013DDAR
R21
100k
Cbst1
2.2nF
InOUT
L3
22uH
Dsw1
SS510C
R22
453k
Ca1
3.3nF
Cb1
56pF
R23
453k
R24
48.9k
+12V
Cout1
22uF
RESET1
R5
10k
C5
100nF
BOOT1
R14
10k
C6
100nF
D17
LED
330
C3
100nF
C4
10uF
U1
ESP32-S3-WROOM-1
EN
3V3
GND
PWR_FLAG
GPIO0
GPIO1
GPIO2
GPIO4
GPIO6
SDA
SCL
GPIO15
TXD0
RXD0
IO17
IO18
USB_D-
USB_D+
IO21
IO35
IO36
IO37
IO38
IO39
IO40
IO41
IO42
IO45
IO46
IO47
IO48
PSRAM
TX
RX
GPIO17
R19
22
R20
22
D-
D+
GPIO38
Relay connector
with low-side
transistor
U2
1-178136-3
Relay connector
+12V
SS510C
D11
D19
LED
R33
1k
Q4
AO3400A
220
R13
R26
10k
3V3 Selector
5VUSB
+12V
3V3
Voltage Sensing Modules
GPIO4
GPIO6
GPIO15
GPIO17
3V3
Vsens1
Vsens2
Vsens3
Vsens4
USB Communication
5VUSB
D+
D-
J2
TX
RX
SDA
GPIO1
SCL
XT30 IN/OUT
connectors
XT30PW-M
CN2
CN4
CN5
CN6
CN7
TP10
TP4
Mounting holes
H4
H3
H2
H1
Sheet: /
File: Good_Boy_Limiter.kicad_sch
Title: GBL (Good Boy Limiter)
Size: A4
Date: 2026-05-01
Rev:
KiCad E.D.A. 9.0.0
Id: 1/5


GND
USB1
USB-TYPE-C-018
GND
R7
5.1K
+5VUSB
D+
D-
R6
5.1K
5VUSBD
R35
330
D18
LED
U4
USBLC6-2SC6
ESD Protection
Sheet: /USB Communication/
File: USB_communication.kicad_sch
Title: GBL
Size: A4
Date: 2026-04-17
Rev:
KiCad E.D.A. 9.0.0
Id: 2/5


3V3
5VUSB
12V
TP1
TP6
TP7
Vin
Cbs1
10nF
U5
SY8120B
L1
10uH
TP5
R8
100k
C7
22uF
R9
22k
C9
100nF
C8
22uF
Sheet: /3V3 Selector/
File: 3V3_selector.kicad_sch
Title:
Size: A4
Date:
Rev:
KiCad E.D.A. 9.0.0
Id: 4/5


3V3
Resistor bridge for sensing capacitor charge
Vsens1
C1 100nF
D12 BAT54S
R2 4.7k
R1 100k
V_Sensing1
U8 B3B-XH-A
3 pins to augment clearence btw VDD and GND to avoid arcs (crippage)
Vsens2
C10 100nF
D13 BAT54S
R4 4.7k
R12 100k
V_Sensing2
U10 B3B-XH-A
Vsens3
C11 100nF
D14 BAT54S
R15 4.7k
R16 100k
V_Sensing3
U11 B3B-XH-A
Vsens4
C12 100nF
D15 BAT54S
R17 4.7k
R18 100k
V_Sensing4
U12 B3B-XH-A
GND
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Title: GBL
Size: A4
Date: 2026-04-20
Rev:
KiCad E.D.A. 9.0.0
Id: 4/5


+48VIN
+48VOUT
VDD
GPIO2
+48VIN
Cin1 100nF
D10 TVS
GPIO2
R25 220
GND Q2 AO3422
R31 143k
TP2 UVLO
R30 10k
R32 210k
R29 10k
TP3 OVLO
Rshort1 0
Ctimer1 1uF
Rpwr1 150k
U3 LM5069NM-2/NOPB
VIN
UVLO
OVLO
TIMER
PWR
PGD
OUT
GATE
SEN
GND
+48VOUT
Cout2 220uF
D2 SS510C
Q1 STB15810
Power_Drain
Rsns1 20m
Sheet: /In-rush controller/
File: inrush_controller.kicad_sch
Title:
Size: A4
Date:
Rev:
KiCad E.D.A. 9.0.0
Id: 5/5

# 15 References


- Texas Instruments LM5069 Datasheet
- Texas Instrument LM5013DDAR Datasheet
- STMicroelectronics STB15810 Datasheet
- Slkor SY8120B Datasheet